TEVA/UTEVA Extraction Chromatography Protocol for Neptunium in High-Level Waste: A Safeguards Technical Document Revised Version 2

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20/05/2026

Abstract

Neptunium-237 (^{237}Np) is one of the most challenging radionuclides in nuclear safeguards and material accountancy due to its long half-life (2.14×10^6 years), complex redox chemistry, and low specific activity. Accurate quantification of Np in high-level radioactive waste (HLW) is essential for verifying declared inventories, detecting potential diversion, and ensuring compliance with IAEA safeguards agreements (INFCIRC/153, INFCIRC/66).

This work presents a technical document for the separation and quantification of neptunium in HLW using modern extraction chromatography resins, specifically the sequential application of TEVA (for Np(IV)) and UTEVA (for Np(VI)). The protocol addresses the critical operational challenges that have historically limited the reliability of Np assays, including:

- Precise redox control to lock Np in a single oxidation state
- Effective masking of ambient fluoride using aluminum nitrate
- Management of uranium co-retention on UTEVA
- Robust uncertainty budgeting following the Guide to the Expression of Uncertainty in Measurement (GUM)
- Full material accountancy traceability with certified yield tracers (Np-239)

The document represents a significant improvement over legacy Dowex 1-X8 anion exchange methods by offering higher decontamination factors, faster kinetics, and better compatibility with modern alpha-spectrometry and ICP-MS workflows.

1. Redox Control Strategy

Neptunium speciation governs extraction chromatography retention. The fundamental safeguard requirement is complete locking of Np into a single oxidation state prior to column loading. Drift between +4 and +6 states manifests as D-value broadening, split peaks in elution profiles, and variable chemical yields.

1.1. Primary Reductant Route (Np(IV) → TEVA)

- **Reagent System:** Ferrous sulfamate combined with ascorbic acid in an HNO_3 matrix serves as the primary reductant for Np(IV)/Pu(IV) valence locking. This replaces $SnCl_2$ to prevent tin precipitation during evaporation, which otherwise degrades alpha-spectroscopy resolution via self-absorption.

- **Acid Window & Kinetics:** 2.5–3 M HNO_3 optimizes tetravalent actinide retention on TEVA resin. Mask ambient fluoride from the HLW matrix with $Al(NO_3)_3$ prior to loading to prevent neutral fluoro-complex formation that would otherwise bypass the quaternary amine active sites. Maintain contact time ≥ 10 min at ambient temperature (20–25°C).

- **Decision Logic:** Select +4/TEVA route when matrix contains high Pu+Am load (>10 µg total actinide), absolute uranium separation is required, or when HF compatibility with downstream alpha electrodeposition is needed.

1.2. Primary Oxidant Route (Np(VI) → UTEVA)

- **Reagent System:** Primary: $NaBiO_3$ (solid, added in excess relative to Np). Fast kinetics, minimal cation addition, clean background. Backup: $(NH_4)_2S_2O_8$ + catalytic Ag^+ (50 ppm). Oxidant strength must exceed +1.4 V vs SHE to drive Np(V)/Np(IV) → Np(VI).

- **Acid Window & Kinetics:** 1–2 M HNO_3 or 0.5–1 M $HClO_4$/HCl mix with $Al(NO_3)_3$ masking for ambient fluoride. Contact time 10–30 min at RT. Avoid HCl >2 M with $NaBiO_3$ to suppress Cl_2 evolution; if using $(NH_4)_2S_2O_8$, maintain ≥3 M H_2SO_4 or HNO_3 for Ag^+ catalyst stability.

- **Decision Logic:** Select +6/UTEVA route for cleaner matrices, lower total actinide load (<5 µg), or when HNO_3 compatibility with downstream ICP-MS or gamma counting is preferred.

1.3. Redox Drift Detection & Correction (Real-Time)

- **Spectroscopic Monitoring:** UV-Vis/NIR absorbance shift at 960 nm (Np(IV) state) vs 1222 nm (Np(VI) state). Appearance of a sharp peak at 980 nm indicates unwanted Np(V) speciation (redox failure). A >15% deviation in target peak integration indicates drift; correct by adding 0.5 mL of appropriate redox reagent and re-equilibrating for 10 min.

- **Chemical Yield Verification:** Spike with certified Np-239 (produced by neutron activation of U-238 or as short-lived tracer). Monitor its 277–296 keV gamma line pre- and post-separation. Recovery <85% signals incomplete redox locking or column breakthrough.

- **Potentiometric Backup:** Use Pt/Ag-AgCl reference pairs for continuous Eh monitoring. Target Eh $>$ +0.3 V (+4 route) or Eh $>$ +1.2 V (+6 route). Drift $> \pm 0.05$ V warrants re-dosing.

1.4. Interfering Species & Suppression

- **Pu(IV)/Am(III):** Pu(IV): Co-extracts strongly with Np(IV) on TEVA. This is suppressed by adding an intermediate wash containing NH_4I (0.1 M) and ascorbic acid to reduce Pu to the un-retained +3 state, washing it from the column prior to Np elution. Trivalent interferents like Am(III) do not retain on TEVA in 3 M HNO_3 and are removed in the initial load effluent.

- **Fe(III), Al(III), Zr/Nb:** Compete for HF complexation; maintain [HF] $\geq$ 0.05 M excess relative to total trivalent/rare earth load (if not masked by Al).

- **Organic Load (>500 mg/L):** Disrupts micellar species. High organic loads must be destroyed via wet ashing (digestion with concentrated HNO_3 and H_2O_2) prior to column loading to ensure isotopic equilibration and prevent violent precipitation/channeling in the resin bed.

2. Extraction Chromatography (TEVA + UTEVA Sequential Method)

2.1. Column Preparation & Preconditioning

- **Resin Choice:** TEVA resin (Aliquat 336 on quaternized acrylic polymer, 50–100 μm bead size). UTEVA resin utilizes *diamyl amylphosphonate* (DAAP) supported on an inert organic polymeric substrate (typically Amberchrom CG-71), not silica. Both are optimized for HLW matrices.

- **Bed Volume:** Minicolumns (2 mL bed) or macro-columns (5–10 mL) depending on expected Np mass (typically 10–100 μg). Pack under gravity or low backpressure (<0.5 bar). Never exceed 1.0 mL/min flow to avoid channeling.

- **Preconditioning Sequence:** Flush with load acid, then equilibrate for 2 bed volumes. Verify baseline pressure drop; >30% increase indicates resin degradation or particulate fouling.

2.2. Sample Loading Conditions

- **Flow rate:** 0.5–1.0 mL/min (gravity preferred for reproducible hydrodynamics). Adjust sample matrix to match chosen redox route acid media ($\pm$0.1 M tolerance). Spike with Np-239 yield monitor prior to loading. Total load volume $\leq$ 10 mL; dilute if TSS $>$ 5 g/L or viscosity $>$ 1.5 cP.

- **Decision point:** If sample contains high organics, perform wet ashing preprocessing before loading. For matrices containing ambient HF, mask with $Al(NO_3)_3$ to free Np(IV)/Np(VI) for resin uptake. Exchange to HNO_3/HCl via evaporation if initial matrix uses >2 M $HClO_4$.

2.3. *Washing Sequences & Rationale*

TEVA Wash Sequence: (Sequential Two-Acid Strip)

- **Load Acid & 3 M HNO_3 (5–8 mL):** Clears the load acid, U(VI), Am(III), and other trivalent fission products from the column. Rationale: $D_{\mathrm{U(VI)}} < 0.1$ under optimized nitric acid conditions. Np(IV), Pu(IV) and Th(IV) remain firmly retained.
- **9 M HCl (10–15 mL):** Targeted Thorium strip. The high chloride concentration forces Th(IV) into the effluent (as it does not form extractable anionic chloro-complexes), while Np(IV) transitions to highly stable anionic chloro-complexes and remains locked on the resin.

Operational Note: The matrix transition from 3 M HNO_3 to 9 M HCl directly on the column may cause minor localized heat or outgassing. Maintain flow rates ≤ 1.0 mL/min to prevent channeling. Ensure PFA and PTFE labware and surrounding hot cell infrastructure can tolerate the brief concentrated chloride exposure.

UTEVA Wash: 1–2 M HNO_3 (10 mL) removes Th/Pu/Am. Acknowledge that U(VI) exhibits exceptionally high affinity for DAAP and will co-retain with Np(VI). If Pu/Am load >10 µg total, insert intermediate wash: 4 M HCl + 0.25 M HF + 0.1 M NH_4I to reduce trivalent actinides that elute early, leaving Np(VI) clean on resin.

Branching Logic: If wash effluent shows >1% of total alpha activity or gamma spike breakthrough, extend wash by 5 mL increments until baseline is reached.

2.4. Elution & Stripping Conditions

- **TEVA Elution (Np(IV))**: 0.1 M HCl or 0.05 M HF (5–8 mL). Acid eluent yields sharper peaks and superior alpha-spectroscopy resolution by preventing colloid formation on electrodeposition. Ethanol is no longer added to the eluent; organics are managed via prior wet ashing.
- **UTEVA Elution (Np(VI))**: 0.01–0.1 M HNO_3 or deionized water + ≥0.05 M HF (6–10 mL). Maintain final eluent [HF] to keep Np in solution during evaporation/drying steps.
- **Branching Decision Points:**
  - D_{Np} < expected range → verify redox state, add fresh oxidant/reductant, re-equilibrate column with 10 mL load acid, reload.
  - Split peaks or tailing → reduce flow rate to ≤0.5 mL/min, increase HF masking in wash/eluent, or switch to TEVA if UTEVA shows excessive uranium co-elution.
- **Recovery Targets:** Chemical yield ≥90% (verified by Np-239 spike). Alpha recovery ≥85%. D-values: TEVA D_{Np} 300–800; UTEVA D_{Np} 150–400 under optimized conditions.

2.5. Fraction Collection & Processing

Collect eluate into PFA or polypropylene vessels (avoid glass for HF compatibility). Evaporate to dryness under mild heat (<90°C) with HNO_3/HF assist if required for electrodeposition. Never exceed 1 M final acid in plating solution to prevent passivation of electrodeposits.

3. Quality Control & Method Validation

3.1. Blanks, Spikes & Matrix-Matched Standards

- **Procedural Blanks:** Eluent-only processed through full column sequence. Target count rate <20% of sample LOQ. If >5%, identify HF leaching from resin or ambient contamination source.
- **Spikes:** Certified Np-237/Np-239 solution (NIST-traceable). Spike at 10–20% of expected sample activity to avoid saturation and ensure linear response in alpha counting.
- **Matrix-Matched CRMs:** IAEA SL-1/SL-2 or laboratory-prepared HLW simulants (simulated fission product load: $UO_2(NO_3)_2$, $Pu(NO_3)_4$, $Am(NO_3)_3$, $Fe(NO_3)_3$, $ZrOCl_2$). Validate annually or after resin batch change.

3.2. Uncertainty Budget (GUM Approach)

To meet IAEA verification standards, the uncertainty budget must strictly follow the Guide to the Expression of Uncertainty in Measurement (GUM).

The final quantification of Np-237 activity concentration requires propagating standard uncertainties across all gravimetric, volumetric, radiometric, and chemical yield parameters. The mathematical model for the specific activity A_{237} (Bq/L or Bq/g) is defined as:

$$A_{237} = \frac{R_a - R_b}{V \cdot \varepsilon \cdot Y \cdot P_\alpha}$$

Where:

- R_a = Gross count rate of the Np-237 region of interest (s^{-1})
- R_b = Background/blank count rate (s^{-1})
- V = Sample mass or volume loaded (g or L)
- ε = Detector counting efficiency
- Y = Chemical yield (derived from Np-239 gamma tracer recovery)
- P_α = Alpha emission probability for the integrated peaks

Law of Propagation of Uncertainty:

The combined standard uncertainty, $u_c(y)$, is calculated using the first-order Taylor series approximation of the measurement equation:

$$u_c^2(y) = \sum \left(\frac{\partial f}{\partial x_i} \right)^2 u^2(x_i)$$

Component Breakdown:

- **Type A Uncertainties (Statistically evaluated):**
  - **Counting Statistics (u_{R_a}, u_{R_b}):** Driven by Poisson statistics. Relative uncertainty scales with the inverse square root of total counts. Extended counting times (up to 72 hours) may be required for low-level safeguards samples to achieve $< 2\%$ relative uncertainty.
  - **Yield Variance (u_Y):** Derived from the counting statistics of the Np-239 277 keV gamma peak pre- and post-separation.

- **Type B Uncertainties (Evaluated by other means):**
  - **Detector Efficiency (u_ε):** Sourced from the calibration certificate of the primary standard source, typically 1.5–3%.
  - **Volumetric/Gravimetric Tolerance (u_V):** Manufacturer specifications for Class A pipettes or analytical balance calibration logs (typically $<$ 0.5%). Assumed as a rectangular or triangular distribution.
  - **Self-Absorption:** Modeled as a bias factor. If electrodeposit thickness exceeds 0.5 mg/cm^2, spectral tailing must be quantified via peak-fitting software to estimate lost counts.

Expanded Uncertainty:

The combined expanded uncertainty is reported with a coverage factor of $k = 2$, representing a 95% level of confidence:

$$U = k \cdot u_c(A_{237})$$

Target combined expanded uncertainty for routine safeguards quantification is $U \leq 8\%$. All uncertainty components, sensitivity coefficients, and degrees of freedom must be documented in laboratory QA logbooks. Runs demonstrating a coverage factor exceeding 10% must be flagged for re-analysis or investigated for systematic column failure.

3.3. Key Performance Indicators

- **Recovery:** 90–110% ($\pm$5% tolerance). Out-of-range indicates redox drift or column overload.
- **Decontamination Factors (DF):** TEVA: $DF_{Th,U} \geq 10^3$, $DF_{Pu,Am} \geq 10^2$. UTEVA: $DF_{Th,U} < 10$ (Uranium will heavily co-retain), $DF_{Pu,Am} \geq 10^2$. Verify by gamma/ICP analysis of wash fractions or alpha spectral deconvolution.
- **Repeatability:** RSD $\leq$ 3% across replicate separations (n=6). Higher variance signals acid concentration drift, flow rate instability, or resin degradation.

3.4. Validation & Inter-Laboratory Comparison

Cross-validate against reference methods: ICP-MS for matrix elements (with mass-shift correction), alpha spectrometry for Np-237 quantification, and gamma spectrometry for Np-239 tracer recovery. Participate in biennial IAEA inter-comparison exercises. Document method validation per IAEA TECDOC-1694 criteria: linearity ($R^2 \geq 0.995$), LOD/LOQ, robustness testing ($\pm$0.1 M acid tolerance).

4. Safety Analysis

4.1. Radiological Hazards

- **Alpha Emission:** Np-237 and progeny pose inhalation/skin absorption risk. Work in HEPA-filtered glovebox or hot cell with continuous air monitoring.

- **Criticality Potential:** Concentrated Np + Pu/U mixtures may approach $k_{\text{eff}} \approx 0.95$ if solvent evaporation is incomplete. Maintain <1% uranium equivalent in final dried residue; use neutron absorbers (borated polyethylene) on storage racks.

4.2. Chemical & Operational Hazards

- **HF Corrosion:** Attacks glass/silica; use PFA, PTFE, or polypropylene exclusively. Monitor HF vapor with acid-ion chromatography strips or electrochemical sensors.

- **Oxidant Handling:** $NaBiO_3$ is a strong oxidizer; avoid contact with organic solvents/dust. Prepare fresh daily; store desiccated at 4°C under inert atmosphere if >24 h use expected.

- **Column Overpressure:** Excessive loading or particulate loading can cause resin bed fracture. Maintain $\Delta P < 0.5$ bar; backflush if pressure spikes.

4.3. *Critical Control Points & Monitoring*

Table 1: Critical Control Points

CCP	Parameter	Acceptance Criteria	Monitoring Frequency
Redox Lock	Eh / UV-Vis ratio	$\pm$0.05 V / <15% drift	Pre-load & post-wash
HF Masking	$[Al(NO_3)_3]$ excess	$\geq$0.05 M equivalent	Post-elution ICP or titration
Column Integrity	Pressure drop, baseline alpha	$\Delta P \leq 0.5$ bar; blank <20% LOQ	Every run
Spike Recovery	Np-239 yield	85–115%	Per sample batch (n$\geq$1)

5. Material Accountancy & Safeguards Integration

5.1. *Traceability Requirements*

Maintain unbroken chain of custody from sample receipt through electrodeposition to alpha counting. Use NIST-traceable spikes, certified reference materials, and lot-controlled resins. Document all reagent preparations, column serial numbers, and operator signatures in bound QA logbooks. Digital backups must be written-once (WORM) for IAEA auditability.

5.2. *Mass Balance Closure*

Calculate final Np-237 total batch inventory (Q) using:

$$Q = A_{237} \cdot V_{\text{batch}}$$

Note: The specific activity (A_{237}) already accounts for chemical yield (Y) and detector efficiency (ε). Acceptable expanded uncertainty: $\leq$ 8% (k=2). Close mass balance within $\pm$10% of declared inventory; investigate drifts via redox monitoring logs or column DF recalibration.

5.3. *Measurement Techniques & Limitations*

- **Alpha Spectrometry:** Primary quantification method for Np-237. Resolves alpha peaks from U/Pu/Am progeny. Self-absorption corrections required if electrodeposit thickness >0.5 mg/cm^2.
- **ICP-MS:** Useful for matrix verification and HF complexation studies; limited sensitivity for actinides without dilution/mass-shift (e.g., $^{93}Zr^+$ on $^{237}Np^+$). Matrix matching essential for <10% relative error.
- **Gamma Spectrometry:** Np-239 spike quantification via 277–296 keV lines. Efficiency calibrated with Co-57/Am-241 standards. Cannot replace alpha for total Np inventory but validates chemical yield independently.

6. Comparison with Legacy HCl Anion Exchange

The Dowex 1-X8 (HCl anion exchange) method relies on formation of $[NpF_6]^{2-}$ or $[NpCl_6]^{2-}$ complexes for retention. While historically foundational, it exhibits several limitations in modern HLW safeguards workflows:

- **Kinetics & Redox Control:** Slower complexation rates; requires prolonged equilibration (15–30 min) and precise acid concentration control ($\pm$0.2 M). Prone to redox drift in fluoride-rich matrices.
- **Selectivity:** Poor DF for trivalent actinides compared to TEVA/UTEVA; co-extraction of
Pu(IV)/Am(III) necessitates complex multi-step washes. Resin fouling by colloidal Zr/Nb fission products is common, causing flow rate decay and tailing.
- **Acid Consumption:** High HCl demand (3–6 M) generates volatile Cl_2 with $NaBiO_3$ oxidation; increases corrosion load on hot cell infrastructure.
- **When Still Used:** Historical archive re-analysis, specific Pu Np separation where TEVA/UTEVA is unavailable, or when HF compatibility with legacy resin systems is mandated. Inferior reproducibility and D-value consistency vs. modern extraction chromatography. The shift to nitric acid media for TEVA loading (2.5–3 M) now provides superior tetravalent retention over HCl anion exchange, while UTEVA's DAAP chemistry offers high throughput at the trade-off of uranium co-retention.

Recommended Summary Table

Table 2: Key operational parameters

Parameter	**TEVA Route (Np(IV))**	**UTEVA Route (Np(VI))**
Active Extractant	Aliquat 336 (quaternary amine)	DAAP (diamyl amylphosphonate)
Redox State	Lock +4 via Ferrous sulfamate + ascorbic acid	Lock +6 via $NaBiO_3$ or $S_2O_8^{2-}/Ag^+$
Acid Media (Load)	2.5–3 M HNO_3 (Mask any ambient HF with $Al(NO_3)_3$)	1–2 M HNO_3 (Mask HF w $Al(NO_3)_3$)
Wash Sequence	3 M HNO_3 (5–8 mL) for U → 9 M HCl (10–15 mL) for Th	1–2 M HNO_3 (10 mL)
Elution	0.1 M HCl or 0.05 M HF (5–8 mL)	0.01–0.1 M HNO_3 or DI H_2O (6–10 mL)
Decision Trigger	High Pu/Am load or abs. req. for U/Np separation	< 5 μg actinides; U co-elution tolerated
Target DF (Th/U)	$\geq 10^3$ (Achieved via the 9 M HCl strip)	< 10 (Uranium will heavily co-retain)
Chemical Yield	90–110% (Np-239 verified)	85–110% (Np-239 verified)

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Mathematical & Metrological Verification Update

- The document provides the general first-order Taylor series approximation for combined standard uncertainty:

$$u_c^2(y) = \sum \left(\frac{\partial f}{\partial x_i} \right)^2 u^2(x_i)$$

- To satisfy the requirement of working out every step without bridging constants, the partial derivatives (sensitivity coefficients, c_i) must be explicitly resolved from the specific activity equation.

- Applying the partial derivatives to A_{237} with respect to each input quantity yields:

- Gross Count Rate (R_a):

$$c_{R_a} = \frac{\partial A_{237}}{\partial R_a} = \frac{1}{V \cdot \epsilon \cdot Y \cdot P_\alpha}$$

- Background Rate (R_b):

$$c_{R_b} = \frac{\partial A_{237}}{\partial R_b} = \frac{-1}{V \cdot \epsilon \cdot Y \cdot P_\alpha}$$

- Volume/Mass (V):

$$c_V = \frac{\partial A_{237}}{\partial V} = -\frac{R_a - R_b}{V^2 \cdot \epsilon \cdot Y \cdot P_\alpha} = -\frac{A_{237}}{V}$$

- Efficiency (ϵ):

$$c_\epsilon = \frac{\partial A_{237}}{\partial \epsilon} = -\frac{A_{237}}{\epsilon}$$

- Chemical Yield (Y):

$$c_Y = \frac{\partial A_{237}}{\partial Y} = -\frac{A_{237}}{Y}$$

- Alpha Emission Probability (P_α):

$$c_{P_\alpha} = \frac{\partial A_{237}}{\partial P_\alpha} = -\frac{A_{237}}{P_\alpha}$$

- Assuming all input quantities are uncorrelated, substituting these sensitivity coefficients back into the GUM summation and dividing by A_{237}^2 yields the relative variance formula. This is the operational equation that should be programmed into the laboratory's calculation matrix:

- $$\frac{u_c^2(A_{237})}{A_{237}^2} = \frac{u^2(R_a) + u^2(R_b)}{(R_a - R_b)^2} + \frac{u^2(V)}{V^2} + \frac{u^2(\epsilon)}{\epsilon^2} + \frac{u^2(Y)}{Y^2} + \frac{u^2(P_\alpha)}{P_\alpha^2}$$

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